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Liquid–liquid extraction of *p*-xylene from their mixtures with alkanes using 1-butyl-1-methylmorpholinium tricyanomethanide and 1-butyl-3-methylimidazolium tricyanomethanide ionic liquids

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ABSTRACT

Liquid – liquid experimental data for four ternary systems {1-butyl-1-methyl-morpholinium tricyanomethanide, [BMMOR][TCM], or 1-butyl-3-methylimidazolium tricyanomethanide, [BMIM][TCM] + *p*-xylene + octane, or decane} were determined at $T = 298.15$ K and atmospheric pressure. The separation of *p*-xylene from alkanes (octane, or decane) is performed by extraction with the tricyanomethanide -based ionic liquids. Good miscibility of *p*-xylene and practical complete immiscibility of alkanes in ionic liquids have been observed. Selectivity, S and solute distribution ratio, β at constant temperature for these mixtures were calculated and discussed. These parameters derived from the experimental equilibrium data were compared with the values for other ionic liquids available in the literature. The NRTL and UNIQUAC models were used to correlate experimental LLE data. Agreement between experimental and correlated data was satisfactory. The influence of the cation structure of the ionic liquid on the phase equilibrium and the separation ability of these ionic liquids was discussed. Additionally, the water content and basic physicochemical properties, such as: density, ρ and dynamic viscosity, η were determined for pure ILs at wide temperature range at ambient pressure.

1. Introduction

Ionic liquids (ILs) due to their unique properties are used in many branches of the chemical industry. From the point of view of this paper, the use of such compounds in liquid extraction of aromatic hydrocarbons (*p*-xylene) from aliphatic hydrocarbons (octane, or decane) is very interesting. Since these hydrocarbons have boiling points in a close range and several combination form azeotropes, separation of aromatic hydrocarbons from mixtures of aliphatic hydrocarbons is a large technological problem and challenge. The conventional processes for this separation is liquid extraction, suitable for aromatic content in the range of 20–65 wt.%, extractive distillation for the interval 65–90% wt. and azeotropic distillation for more than 90% aromatic content in the mixture [1,2]. Typical solvents used in industry for separation in these systems are: sulfolane [3–5], *N*-methyl-2-pyrrolidinone (NMP) [6,7], *N*-formyl-morpholine [8,9], ethylene glycols [10,11], or propylene carbonate [12].

The use of such solvents entail high costs resulting additional steps to separate the solvent from the two phases (extract and raffinate) and its purification. In addition, these compounds are harmful to the environment due to its high toxicity, flammability and above all high volatility. It is estimated that more than twenty million tons of volatile organic compounds is discharged into the atmosphere each year as a result of the processes related to the chemical processing [13]. These emissions have been associated with the negative effects of the chemical industry on the environment such as: global climate change, air pollution, and human consequences of this disease.

Ionic liquids, as practically non-volatile compounds are seen as the ideal replacement for conventional volatile organic solvents. In the face of numerous problems with the adverse effects of the chemical industry on the environment, it becomes necessary to design technologies using new eco-friendly compounds, hence the motivation in many institutions in the world, to conduct research on the potential use of ionic liquids in many industries. Intensive research related to the possibility of separating mixtures of aromatic hydrocarbons from aliphatic using ionic liquids are conducted by many research groups. Currently, different studies about the separation of aromatic hydrocarbons from alkane using ionic

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liquids as a solvent were published [14–36]. Among them, only a small part concerns the applicability of ionic liquids in separation of xylenes from aliphatic hydrocarbon mixture. This work will provide a valuable supplement databases in the field.

ILs with tricyanomethanide anion, [TCM]- are chemically and thermally stable. The use of this anion can obtain high values of selectivity in the process extraction of aromatic from aliphatic compounds. This group of ionic liquids with polar [TCM]- anion have been carefully studied in my laboratory in liquid – liquid binary systems {IL + alkanes, or alcohol, or water} [37,38], ternary systems {IL + aromatic sulphure compounds + alkanes} [39–41], activity coefficients at infinite dilution [42–44] and in model fuels [45,46] with very good results of the extraction of aromatic sulfur compounds. The best results, high S and β in ternary system were obtained with 1-ethyl-3-methylimidazolium tricyanomethanide, [EMIM][TCM] [40] as well as for 1-butyl-1-methylpyrrolidinium tricyanomethanide, [BMPYR][TCM] [44], or 1-butyl-3-methylimidazolium tricyanomethanide, [BMIM][TCM] [45] for extraction of thiophene from model fuels. The extraction of aromatic sulfur compounds from model fuels was 70–95 mass percent [44,45].

The first step in determining the suitability of the [TCM]- based ionic liquid to effective separation xylenes from aliphatic hydrocarbons was calculated of selectivity, S_{12}^{∞} and capacity, k_2^{∞} at infinite dilution from activity coefficients at infinite dilution, using: [BMIM][TCM] [44], [BMMOR][TCM] [42], [BMPYR][TCM] [43] and [BMPy][TCM] [44]. The selectivity, S_{12}^{∞} and capacity, k_2^{∞} at infinite dilution for the separation of octane/p-xylene at $T = 318.15$ K obtained for IL tested in this work are: $S_{12}^{\infty} = 35.8$ and $k_2^{\infty} = 0.29$ for [BMMOR][TCM] and $S_{12}^{\infty} = 26.0$ and $k_2^{\infty} = 0.38$ for [BMIM][TCM]. In comparison with other ionic liquids, these values are quite high [47–50].

In this work, liquid – liquid equilibrium data have been measure for the ternary systems containing 1-butyl-1-methylmorpholinium tricyanomethanide, [BMMOR][TCM], 1-butyl-3-methylimidazolium tricyanomethanide, [BMIM][TCM], p-xylene and octane, or decane at $T = 298.15$ K at atmospheric pressure. From the experimental data, selectivity (S) and solute distribution ratio (β) were determined and used to evaluate the suitability of the tested ILs as a solvent in the liquid extraction process. The experimental results were compared with those available in the opened literature for others ILs and sulfolane as a solvent.

2. Experimental

2.1. Chemicals

1-Butyl-1-methylmorpholinium tricyanomethanide, [BMMOR][TCM] and 1-butyl-3-methylimidazolium tricyanomethanide, [BMIM][TCM] were supplied by IoLiTec GmbH, Germany and the structures of the investigated ionic liquids are presented in Table 1. Immediately before the experiments, ILs were dried for 48 h in a Vacuum Drying Ovens (Binder, model VD 23) at a temperature $T = 373$ K and under reduce pressure obtained by vacuum pomp (Vacuubrand RZ 6). This is our standard procedure of drying ILs. The basic thermophysical characterization including DSC and TG/DTA technique of the tested ILs was presented earlier [38]. The rest of the chemicals used in this experiment are listed in Table 2 and they were stored over freshly activated molecular sieves of type 5 Å (0.5 nm, Typ 522, Perlform, Carl Roth). Purity of the solvents was checked on gas chromatography (GC).

2.2. Apparatus and procedure

The water content of pure ILs was tested using Karl-Fisher titration method and the result was less than 740 ppm for [BMMOR][TCM] and 660 ppm for [BMIM][TCM]. Density of pure ILs was measured using an Anton Paar 4500 M densimeter and the measurement error was better than $\pm 1 \times 10^{-4}$ g·cm^{-3}. Experimental and literature values of the density of pure ILs are collected in Table 3 and presented in Fig. 1. These methods and apparatus are described in previous papers [40,41].

Viscosity measurements were carried out using an Anton Paar GmbH AMVn (Graz, Austria) programmable viscometer, with a nominal uncertainty of ±5.0% and reproducibility <0.05% for viscosities from (0.3–2500) mPa·s. Temperature was controlled internally with a precision of ±0.01 K in a range from $T = (288.15–368.15)$ K. In the measured viscosity range the diameter of the capillary was 1.8 mm, 3.0 mm and 4.0 mm and the diameter of the steal ball was 1.5 mm, 2.5 mm, 3.0 mm for the following viscosity range: (2.5–70) mPa·s, (20–230) mPa·s and (80–2500) mPa·s, respectively. Experimental and literature values of the dynamic viscosity are presented in Table 3 and Fig. 2.

Mettler Toledo AB204-S balance was used to carry out all weighing during the experiment. Measurement error of weighting is $\pm 1 \times 10^{-4}$ g.

The detail description of experimental procedure for the determination of phase composition in ternary system has been presented in our earlier paper [40]. The gas chromatograph PerkinElmer Clarus 580 GC with FID detector and TotalChrom Workstation software was used for the analysis phase. The measurements were carried out three times, and the average value was calculated. Details of the instrumental settings are given in Table 4. The experimental data for LLE in ternary systems, as well as the values of selectivity and solute distribution ratio are presented in Table 5 and Figs. 3–8.

3. Data correlation

The correlation of experimental data for four ternary systems discussed here was performed using two thermodynamic models, such as NRTL (non-random two liquid) and UNIQUAC (universal quasichemical equation) developed by Renon and Prausnitz [51] and Abrams and Prausnitz [52], respectively. The equations of these models and algorithms used in the calculation can be found in Wales book [53]. The objective function $F(P)$, was used to minimize the difference between the experimental and calculated compositions:

$$F(P) = \sum_{i=1}^{n}\left[x_2^{\mathrm{I\,exp}} - x_2^{\mathrm{I\,calc}}(PT)\right]^2 + \left[x_3^{\mathrm{I\,exp}} - x_3^{\mathrm{I\,calc}}(PT)\right]^2 + \left[x_2^{\mathrm{II\,exp}} - x_2^{\mathrm{II\,calc}}(PT)\right]^2 + \left[x_3^{\mathrm{II\,exp}} - x_3^{\mathrm{II\,calc}}(PT)\right]^2 \quad (1)$$

where: P is the set of parameters vector, n is the number of experimental points, $x_2^{\mathrm{I\,exp}}$, $x_3^{\mathrm{I\,exp}}$ and $x_2^{\mathrm{I\,calc}}(PT)$, $x_3^{\mathrm{I\,calc}}(PT)$ are the experimental and calculated mole fractions of one phase and $x_2^{\mathrm{II\,exp}}$, $x_3^{\mathrm{II\,exp}}$ and $x_2^{\mathrm{II\,calc}}(PT)$, $x_3^{\mathrm{II\,calc}}(PT)$ are the experimental and calculated mole fractions of the second phase.

In this work, for the NRTL model, parameter, α_{ij}, was set at a value of 0.2.

For the ionic liquids, the pure component structural parameters R_i (volume) and Q_i (surface area) in the UNIQUAC equation were estimated from the experimental molar volume at $T = 298.15$ K, by using the following correlation suggested by Hofman and Nagata [54]:

$$R_i = 0.029281\, V_{\mathrm{m}i}$$

Table 1
Description of the investigated ILs.

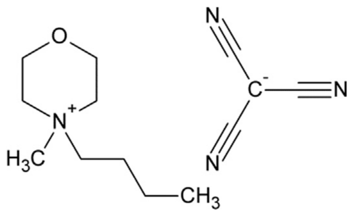
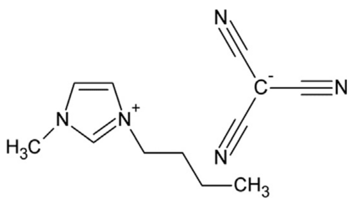
Structure	Abbreviation	Name
	[BMMOR][TCM]	1-Butyl-1-methylmorpholinium tricyanomethanide
	[BMIM][TCM]	1-Butyl-3-methylimidazolium tricyanomethanide

Table 2
List of components used in this work.

Chemical name, CAS No.	Source	Initial mass fraction purity	Purification method	Final mass fraction purity	Analysis method
[BMMOR][TCM]	Io-li-tec	≥0.97	Vacuum heating	–	Karl - Fischer
[BMIM][TCM]	Io-li-tec	≥0.98	Vacuum heating	–	Karl - Fischer
Octane 111-65-9	Sigma Aldrich	≥0.99	Molecular sieves	0.9991	GC analysis, Karl - Fischer
Decane 124-18-5	Sigma Aldrich	≥0.99	Molecular sieves	0.9988	GC analysis, Karl - Fischer
$p-$Xylene 106-42-3	Sigma Aldrich	≥0.99	Molecular sieves	0.9982	GC analysis, Karl - Fischer
$1-$Propanol 71-23-8	Sigma Aldrich	≥0.99	Molecular sieves	0.9987	GC analysis, Karl - Fischer
Acetone 67-64-1	POCH	≥0.99	Molecular sieves	0.9997	GC analysis, Karl - Fischer

Table 3
The experimental and density (ρ) and dynamic viscosity (η), as a function of temperature at $p = 0.1$ MPa for investigated ionic liquids[a].

T (K)	ρ (g cm^{-3})	η (mPa s)
[BMMOR][TCM]		
288.15		500.4
298.15		225.5
308.15		118.4
318.15		69.89
328.15		44.14
338.15		30.31
348.15		21.90
358.15		16.33
368.15		12.77
[BMIM][TCM]		
288.15	1.0536	43.68
298.15	1.0468	27.78
308.15	1.0400	19.22
318.15	1.0334	12.93
328.15	1.0267	9.98
338.15	1.0201	7.78
348.15	1.0136	6.29
358.15	1.0072	5.18

[a] Standard uncertainties u are $u(T) = 0.01$ K and $u(p) = 0.5$ kPa for density and dynamic viscosity measurements, $u(\rho) = 10^{-4}$ g · cm^{-3}, $u_r(\eta) = \pm$ 3%.

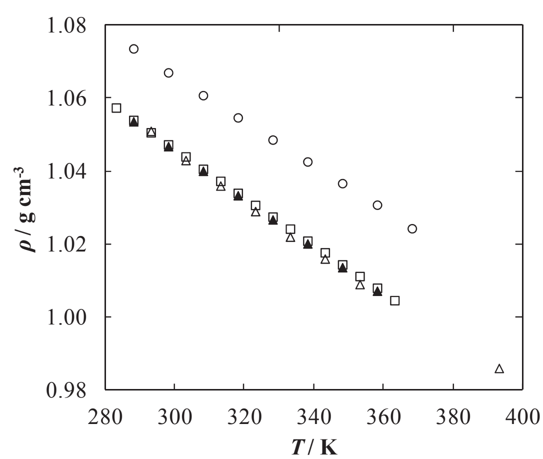


Fig. 1. Density as a function of temperature for the ionic liquids [BMMOR][TCM] (○) [42], [BMIM][TCM] (▲), [BMIM][TCM] (△) [56], [BMIM][TCM] (□) [57].

$$Q_i = \frac{(Z-2)R_i}{Z} - \frac{2(1-l_i)}{Z}$$

where V_{mi} is the molar volume of pure component i, Z is the coordination number, assumed equal to 10 and l_i is the bulk factor which was equated to zero for chainlike molecules and equated to

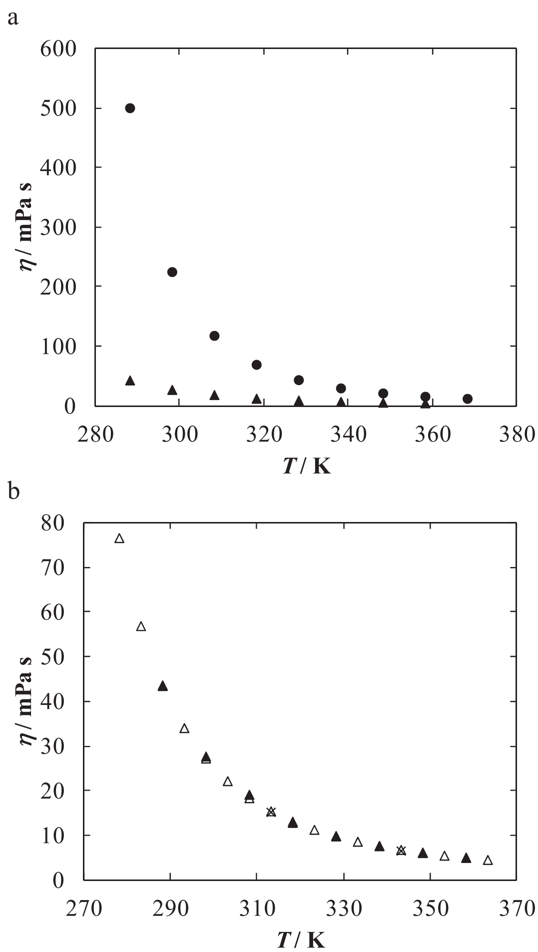


Fig. 2. (a) Dynamic viscosity as a function of temperature for the ionic liquids [BMMOR][TCM] (●), [BMIM][TCM] (▲). (b) Experimental (▲) and literature data (△) [57], (×) [58] of dynamic viscosity as a function of temperature for the ionic liquid [BMIM][TCM].

Table 4
Operational conditions for GC.

Element	Characteristic	Description
Columns	Type	Elit-5 PerkinElmer DB-5 (5% diphenyl/95% dimethyl polysiloxane), length 30 m, inner diameter 0.53 mm, film thickness: 1.5 μm Elite-Wax PerkinElmer, length 30 m, inner diameter 0.53 mm, film thickness: 1.0 μm
	Flow	$5\ cm^3 \cdot min^{-1}$
	Carrier gas	Helium
Oven	Temperature	373.15 K
Injector	Injection volume	$0.0001\ cm^3$
	Split ratio	3:1
	Temperature	423 K
Detector	Type	Flame ionization detector (FID)
	Temperature	493 K

unity for ring molecules. The parameters for $p-$xylene, octane and decane were taken from literature [55]. All parameters R and Q are listed in Table 6.

The correlated parameters obtained using the NTRL and UNIQUAC models along with the root mean square deviations (RMSD) are given in Table 7. The RMSD values, which can be taken as a measure of the precision of the correlation were calculated according the equation:

$$RMSD = \left(\sum_i \sum_l \sum_m \left[x_{ilm}^{\exp} - x_{ilm}^{\exp} \right]^2 \Big/ 6k \right)^{1/2} \quad (2)$$

where x is the mole fraction; the subscript i, l and m provide a designation for the component, phase and the tie-line, respectively. The value k designates the number of tie-lines. The compositions calculated from the correlations are included in Figs. 3–6. The correlation results, obtained for the four tested systems were satisfactory. The experimental and calculated LLE values agrees relatively well. As can be seen from Table 7, the correlation

obtained with the NRTL model is significantly better than that obtained with the UNIQUAC model. The RMSD values are between 0.0036 for {[BMIM][TCM] (1) + p-xylene (2) + octane (3)} and 0.0060 for {[BMMOR][TCM] (1) + p-xylene (2) + octane (3)} ternary systems, while for UNIQUAC model, the values of RMSD are between 0.0057 for {[BMMOR][TCM] (1) + p-xylene (2) + decane (3)} and 0.0077 for {[BMIM][TCM] (1) + p-xylene (2) + octane (3)}.

4. Results and discussion

In order to design a technological processes, the basic physico-chemical properties of pure ionic liquids are very important. From this reason, the density and dynamic viscosity of two ILs that is: [BMIM][TCM] and [BMMOR][TCM] have been measured. Density of [BMMOR][TCM] was measured in our laboratory and presented in previous work [42].

Experimental results are collected in Table 3. The comparison of the density and viscosity of the tested ILs with the literature data are presented in Figs. 1–2 [56–58].

In Table 5 the experimental LLE tie-line for the investigated ternary systems {IL (1) + p-xylene + octane, or decane (3)} at $T = 298.15$ K at ambient pressure are presented. Graphical representation of the experimental results in the form of LLE phase diagrams including NRTL correlation are presented in Figs. 3–6.

The solubility of aliphatic hydrocarbons in the tested ionic liquids was very low. The complete liquid miscibility occurs from IL mole fraction: $x_1 = 0.992$, $x_1 = 0.997$ and $x_1 = 0.982$, $x_1 = 0.991$ for [BMMOR][TCM] with octane, decane and [BMIM][TCM] with octane, decane, respectively. This is due to the polar nature of IL and very weak interaction with an aliphatic hydrocarbons. The lower solubility of alkanes is in morpholinium-based IL and is slightly larger in the imidazolium-based IL.

In the binary {IL (1) + p-xylene (2)} system, the complete miscibility in the liquid phase occurs from mole fraction $x_1 = 0.620$ and $x_1 = 0.514$ to $x_1 = 1$ for [BMMOR][TCM] and [BMIM][TCM], respectively.

Table 5
Experimental mole fractions of the liquid-liquid extraction, selectivity, (S) and solute distribution ratios, (β) for ternary systems {[BMMOR][TCM] or [BMIM][TCM] (1) + p-xylene (2) + octane, or decane (3)} at T = 298.15 K and p = 0.1 MPa[a].

Hydrocarbon − rich phase		IL − rich phase		S	β
x_2^I	x_3^I	x_2^{II}	x_3^{II}		
[BMMOR][TCM] (1) + p-xylene (2) + octane (3)					
0.000	1.000	0.000	0.008		
0.053	0.926	0.027	0.006	78,6	0.51
0.149	0.846	0.065	0.008	46,1	0.44
0.290	0.710	0.114	0.007	39.9	0.39
0.384	0.616	0.144	0.007	33.0	0.38
0.488	0.512	0.183	0.006	32.0	0.38
0.563	0.437	0.203	0.005	31.5	0.36
0.638	0.362	0.233	0.005	26.4	0.37
0.713	0.287	0.258	0.005	20.8	0.36
0.807	0.193	0.295	0.005	14.1	0.37
0.897	0.092	0.357	0.007	5.2	0.40
1.000	0.000	0.380	0.000		0.38
[BMMOR][TCM] (1) + p-xylene (2) + decane (3)					
0.000	1.000	0.000	0.003		
0.145	0.855	0.053	0.002	156.3	0.37
0.419	0.581	0.141	0.002	97.8	0.34
0.549	0.451	0.173	0.002	71.1	0.32
0.636	0.364	0.228	0.002	65.2	0.36
0.695	0.305	0.252	0.002	55.3	0.36
0.754	0.246	0.268	0.002	43.7	0.36
0.810	0.190	0.293	0.002	34.4	0.36
0.867	0.133	0.320	0.002	24.5	0.37
0.914	0.086	0.340	0.002	16.0	0.37
1.000	0.000	0.380	0.000		0.38
[BMMIM][TCM] (1) + p-xylene (2) + octane (3)					
0.000	1.000	0.000	0.018		
0.037	0.963	0.021	0.015	36.4	0.57
0.120	0.880	0.065	0.017	28.0	0.54
0.297	0.703	0.155	0.014	26.2	0.52
0.347	0.653	0.174	0.013	25.2	0.50
0.457	0.543	0.225	0.013	20.6	0.49
0.551	0.449	0.277	0.014	16.1	0.50
0.632	0.368	0.306	0.010	17.8	0.48
0.718	0.282	0.348	0.010	13.7	0.48
0.802	0.198	0.382	0.007	13.5	0.48
0.883	0.117	0.428	0.005	11.3	0.48
1.000	0.000	0.486	0.000		0.49
[BMMIM][TCM] (1) + p-xylene (2) + decane (3)					
0.000	1.000	0.000	0.009		
0.075	0.925	0.036	0.008	55.5	0.48
0.158	0.842	0.077	0.008	51.3	0.49
0.231	0.769	0.106	0.007	50.4	0.46
0.382	0.618	0.185	0.008	37.4	0.48
0.484	0.516	0.213	0.006	37.8	0.44
0.576	0.424	0.272	0.005	40.0	0.47
0.642	0.358	0.305	0.006	28.3	0.48
0.753	0.247	0.358	0.004	29.4	0.48
0.824	0.176	0.397	0.004	21.2	0.48
0.884	0.116	0.430	0.003	18.8	0.49
1.000	0.000	0.486	0.000		0.49

[a] Standard uncertainties u are: $u(x) = 0.003$ for compositions of the hydrocarbon-rich phase, $u(x) = 0.005$ for compositions of IL-rich phase, $u(T) = 0.02$ K, $u(p) = 0.5$ kPa.

The complete miscibility is observed in the {p-xylene (2) + octane, or decane (3)} binary system.

From separation point of view, two factors that is: selectivity, S and solute distribution ratio, β are the most important. These parameters are used to determine the potential ability of the tested ionic liquids for extraction of aromatic from aliphatic hydrocarbons. These parameters reported in Table 5, together with the liquid - liquid data were calculated from experimental data according to equations:

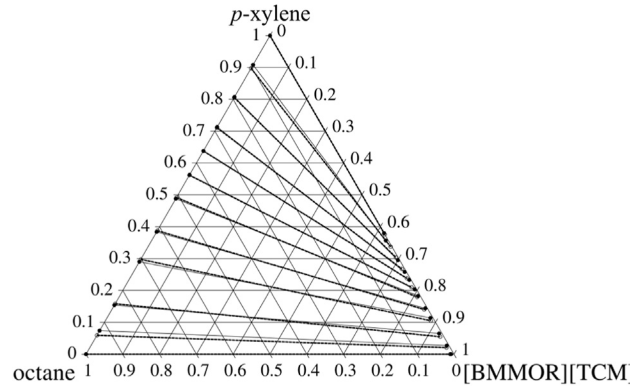


Fig. 3. Experimental tie-lines (●, solid lines) for the LLE of the ternary system {[BMMOR][TCM] (1) + p-xylene (2) + octane (3)} at T = 298.15 K and p = 0.1 MPa. The corresponding tie-lines correlated by means of the NRTL equation (○, dotted lines).

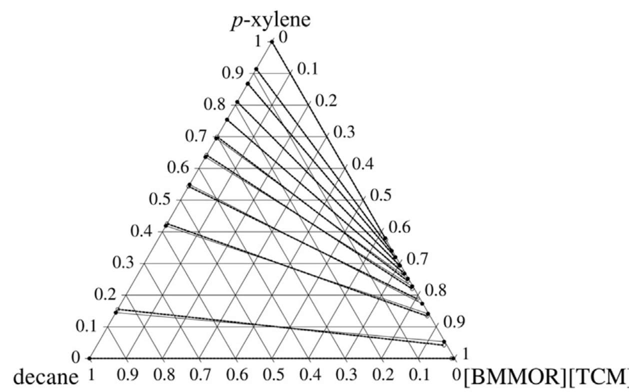


Fig. 4. Experimental tie-lines (●, solid lines) for the LLE of the ternary system {[BMMOR][TCM] (1) + p-xylene (2) + decane (3)} at T = 298.15 K and p = 0.1 MPa. The corresponding tie-lines correlated by means of the NRTL equation (○, dotted lines).

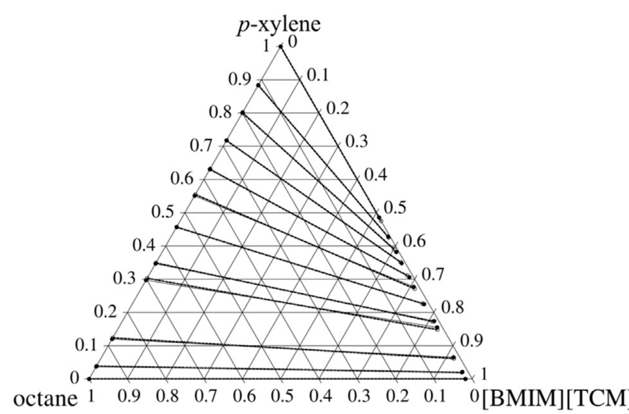


Fig. 5. Experimental tie-lines (●, solid lines) for the LLE of the ternary system {[BMIM][TCM] (1) + p-xylene (2) + octane (3)} at T = 298.15 K and p = 0.1 MPa. The corresponding tie-lines correlated by means of the NRTL equation (○, dotted lines).

$$S = \frac{x_2^{II} x_3^{I}}{x_2^{I} x_3^{II}} \tag{3}$$

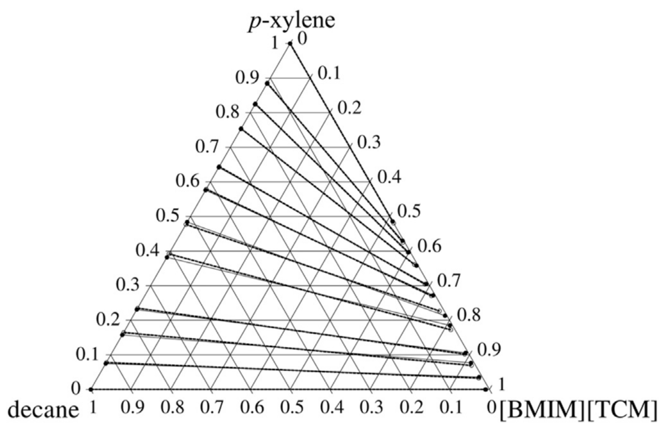


Fig. 6. Experimental tie-lines (●, solid lines) for the LLE of the ternary system {[BMIM][TCM] (1) + *p*-xylene (2) + decane (3)} at $T = 298.15$ K and $p = 0.1$ MPa. The corresponding tie-lines correlated by means of the NRTL equation (○, dotted lines).

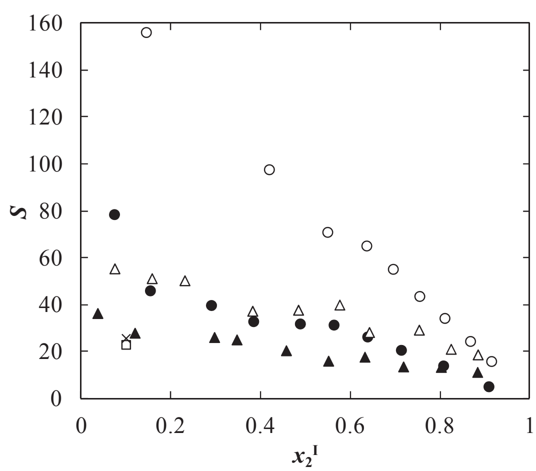


Fig. 7. Plot of the selectivity as a function of the mole fraction of solute in the hydrocarbon − rich phase for the ternary systems: {[BMMOR][TCM] (1) + *p*-xylene (2) + octane (3)} (●), {[BMMOR][TCM] (1) + *p*-xylene (2) + decane (3)} (○), {[BMIM][TCM] (1) + *p*-xylene (2) + octane (3)} (▲), {[BMIM][TCM] (1) + *p*-xylene (2) + decane (3)} (△), {[BMIM][TCM] (1) + *p*-xylene (2) + octane (3)} at $T = 313.15$ K (□) [2], {[B^{3}MPy][TCM] (1) + *p*-xylene (2) + octane (3)} at $T = 313.15$ K (+) [2].

$$\beta = \frac{x_2^{II}}{x_2^{I}} \tag{4}$$

where: *x* is the mole fraction; superscripts I and II refer to the hydrocarbon-rich phase (raffinate) and ionic liquid-rich phase (extract), respectively. Subscripts 2 and 3 refer to *p*-xylene and octane, or decane, respectively. Fig. 7 shows the values of selectivity as a function of the mole fraction of solute in the hydrocarbon rich phase.

It was observed that in each of the tested system the values of *S* decrease with increases of the mole fraction of *p*-xylene in the raffinate. For ternary system with octane, the highest values of *S* that are: 60 and 30 were observed for the mole fraction of octane in alkane-rich phase $x_2^{I} = 0.1$ for: [BMMOR][TCM] and [BMIM][TCM], respectively. With increase of the alkyl chain length in the aliphatic hydrocarbons, the interaction with IL decreases resulting in an increase of the values of *S*. The values of *S* for {IL (1) + *p*-xylene (2) + octane (3)} is almost three times lower than that for {IL (1) + *p*-xylene (2) + decane (3)} for [BMMOR][TCM] and almost two times lower for [BMIM][TCM]. The values of *S* for {[BMMOR][TCM] (1) + *p*-xylene (2) + decane (3)} ternary system are three times higher than that for {[BMIM][TCM] (1) + *p*-xylene (2) + decane (3)} ternary system. This is the result of better solubility of *p*-xylene in the ionic liquid - rich phase. The high value of *S* is mainly due to the practically complete immiscibility of aliphatic hydrocarbons in the tested ionic liquids.

Regarding the *S* values, a comparative study between experimental and literature data [2] for the system {IL (1) + *p*-xylene (2) + octane (3)} was done and presented shown in Fig. 7. The selectivity for [BMMOR][TCM] from this work is two times higher than that for [BMIM][TCM] and [B^{3}MPy][TCM] at higher temperature $T = 313.15$ K. The selectivity for [BMIM][TCM] at $T = 298.15$ K from this work is slightly higher than for [BMIM][TCM] at $T = 313.15$ K. This is the result of better solubility of octane in ILs at higher temperature.

Solute distribution ratio, β is a quantitative measure to indicate how a solute is distributed between the extract and raffinate phases. A high value of distribution coefficient is desirable as this means lower solvent to feed ratio and lower operating cost. From experimental results, the value of β were determined and presented as a function of the mole fraction of solute in the raffinate for the tested binary mixtures at Fig. 8. In each cases, the values of β are lower than unity and the dependence of the values of β as a function of the mole fraction of solute in the raffinate. An increase of the alkyl chain length in the aliphatic hydrocarbon did not significantly influence the value of the β. The values of the solute distribution ratio is similar for the tested ILs and is equal to 0.4 and 0.5 for {[BMMOR][TCM] (1) + *p*-xylene (2) + octane, or decane (3)} and {[BMIM][TCM] (1) + *p*-xylene (2) + octane, or decane (3)},

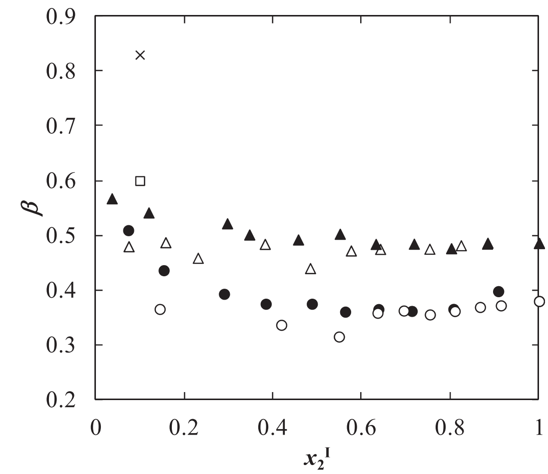


Fig. 8. Plot of the solute distribution ratio as a function of the mole fraction of solute in the hydrocarbon − rich phase for the ternary systems: {[BMMOR][TCM] (1) + *p*-xylene (2) + octane (3)} (●), {[BMMOR][TCM] (1) + *p*-xylene (2) + decane (3)} (○), {[BMIM][TCM] (1) + *p*-xylene (2) + octane (3)} (▲), {[BMIM][TCM] (1) + *p*-xylene (2) + decane (3)} (△),{[BMIM][TCM] (1) + *p*-xylene (2) + octane (3)} at $T = 313.15$ K (□) [2], {[B^{3}MPy][TCM] (1) + *p*-xylene (2) + octane (3)} at $T = 313.15$ K (+) [2].

Table 6
UNIQUAC structural parameters.

Component	*R*	*Q*
[BMMOR][TCM]	6.814	5.451
[BMIM][TCM]	6.413	4.914
p-xylene	4.658	3.536
octane	5.849	4.936
decane	7.198	6.016

Table 7
NRTL and UNIQUAC binary interaction parameters and RMSD (σ_x) for ternary systems {[BMMOR][TCM] or [BMIM][TCM] (1) + p-xylene (2) + octane, or decane (3)} at T = 298.15 K, p = 0.1 MPa.

Components i-j	Parameters					
	NRTL[a]			UNIQUAC		
	g_{ij}-g_{jj} (J mol^{-1})	g_{ji}-g_{ii} (J mol^{-1})	RMSD σ_x	u_{ij}-u_{jj} (J mol^{-1})	u_{ji}-u_{ii} (J mol^{-1})	RMSD σ_x
[BMMOR][TCM] (1) + p-xylene (2) + octane (3)						
1–2	−529.92	27486.59	0.0060	−854.50	6906.70	0.0067
1–3	7175.97	15822.81		131.26	5557.15	
2–3	2390.11	−2182.69		3060.26	−936.35	
[BMMOR][TCM] (1) + p-xylene (2) + decane (3)						
1–2	−24.20	29626.51	0.0053	−283.56	5034.35	0.0057
1–3	7901.14	18346.53		679.08	18665.93	
2–3	4828.98	−4021.61		5866.08	−806.69	
[BMMIM][TCM] (1) + p-xylene (2) + octane (3)						
1–2	−1497.96	29744.49	0.0036	−1717.09	21441.21	0.0077
1–3	6195.22	17492.00		−361.71	27118.87	
2–3	3328.62	−3140.23		4752.16	−711.76	
[BMMIM][TCM] (1) + p-xylene (2) + decane (3)						
1–2	−768.67	32960.91	0.0050	−1466.13	6516.84	0.0064
1–3	7180.24	15812.27		−481.85	7992.88	
2–3	3766.34	−3507.96		4614.40	−727.66	

[a] Parameter α_{ij} = 0.2.

respectively. The value of β for the tested IL are optimistic from separation point of view. For comparison, the values of β for {[B^{3}MPy][TCM] (1) + p-xylene (2) + octane (3)} at higher temperature T = 313.15 K for the mole fraction of octane in hydrocarbon-rich phase x_2^{I} = 0.1 is 0.83 and for {[BMIM][TCM] (1) + p-xylene (2) + octane (3)} at T = 313.15 K for x_2^{I} = 0.1 is 0.60 [2]. This is the result of better solubility of p - xylene in the ILs at higher temperature.

5. Conclusions

The main aim of this work is determine the influence of ILs cation and the alkyl chain length in the hydrocarbons on the mutual solubility in ternary systems. Liquid - liquid phase equilibria were carried out to investigate the ability of the tested ionic liquids to selectivity extract aromatic from aliphatic hydrocarbons mixture. Four ternary systems {tricyanomethanide-based IL (1) + p-xylene (2) + octane, or decane (3)} were determined using GC for the composition analysis at temperature T = 298.15 K and at atmospheric pressure. The comparison of experimental results with other systems studied gave us an opportunity to determine the impact of the cation structure and the alkyl chain length on the phase equilibria and the separation ability of the tested ionic liquids. From separation point of view, two factors selectivity, S and solute distribution ratio, β were calculated from experimental data. The highest value of S was determined for {[BMMOR][TCM] (1) + p-xylene (2) + decane (3)}.

Despite the high values of S, low values of β between 0.32 and 0.57 were obtained. It is not optimistic from separation technology on the industrial scale because of enormous requirement of solvent.

It was observed that with increase of the alkyl chain length in the aliphatic hydrocarbon, the value of selectivity incereases. The experimental data have been correlated using NRTL and UNIQUAC equations. Additionally, the basic physicochemical properties that is: density and dynamic viscosity have been determined over a wide temperature range and at atmospheric pressure.

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